2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.204 Software Design

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a Describe two event-driven models of control in system architecture design. **[2 marks]**

1.b Use a labelled diagram to illustrate the service-oriented architecture style. Identify two advantages and two disadvantages of the service-oriented architecture style.

[5 marks]

1.c Suggest an appropriate architectural style for each of the following systems and provide your justification:

i. A robot football game where two autonomous robot teams play football. In addition to a movement control system, each robot has a vision system to locate the ball, its teammates and the opposition goalpost **[2 mark]**

ii. A computer-controlled video conferencing system that allows video, audio and computer data to be visible to participants in different locations at the same time

[2 mark]

1.d You have been asked to develop a smart phone application called *myGrocer* for automating supermarket purchases and checkouts. *myGrocer* users must register their credit card details with the supermarket before using the application. The application shall:

i. Scan purchases and add them to a shopping cart on the customer's phone.

ii. Automate the checkout of purchases. As the customer leaves the store, an item sensor shall detect the customer's exit and automatically tally his or her purchases. The item sensor shall alert the store security system if the customer has unscanned items. The system shall take payment via the customer's pre-registered credit card.

iii. Check purchases for compliance with the customer's nutritional profile as the purchases are scanned and alert the customer to accept or decline the purchase.

iv. Display a list of items on 'special offer' that satisfy the customer's nutritional profile" as the customer walks around the supermarket. The smart phone shall obtain the list of items from the supermarket item database.

In order to use the functionality described in (iii), the customer shall be required to enter his/her nutritional profile into the system using a web browser prior to shopping.

Using the **ball and socket** notation, to design a software architecture for *myGrocer* based on an appropriate architecture style. Your design should identify *myGrocer's* main sub-systems and their interfaces. Using a table, list the interfaces, and the interfaces' main operations, inputs and outputs.

[14 marks]

Total 25 marks

Question 2

2.a Describe three graphical models that may be produced as part of architectural design

[3 marks]

2.b *Online Books* wishes to develop an online selling capability via the Web. The website which implements this capability should enable a customer to browse the details of a book, order a book and register as a recipient of emails which will carry details of special offers, new publications and reviews.

New customers will be prompted to register before they can order books. Registration details will include the customer's name, email address and postal address. The system will maintain a virtual shopping cart to store the customer's shopping. As the shopping cart is filled the cumulative cost of the purchases will be displayed to the customer.

If stocks in the warehouse cannot satisfy an order, a back order will be raised and the customer notified. A back order is retailer's order for a product which is temporarily out of stock with the supplier. Satisfied orders will be shipped as *Standard* (i.e. delivered within 3 days) or *Next Day* delivery.

The website will interface with a stock control system, The stock control system will handle the process of ordering new books, updating the stocks when books are bought and reordering book stocks. Each stock item has an associated order line, which contains information about what book is to be ordered and the quantity. Books are automatically reordered when their stocks fall below preset levels.

Draw a sequence diagram showing the process of ordering a book.

[8 marks]

2.c Design a UML class diagram for the system showing the object classes that make up the *Online Books* system, their attributes and operations. Indicate the relationships between the classes and their multiplicity.

[14 marks]

Total 25 marks

Question 3

3.a Define the term 'software requirement'

[1 mark]

3.b State clearly the difference between a functional and a non-functional requirement

[1 mark]

3.c The following are example requirements for a social networking site (SNS). In each case, identify one reason why the requirement is poorly specified. Then, in each case, restate the requirement to improve it based on the problem you have identified.

[6 marks]

i. The SNS shall be easy to use

ii. If the user double clicks on the 'add friend' button, the screen changes to ask the user to enter the name of a friend to add

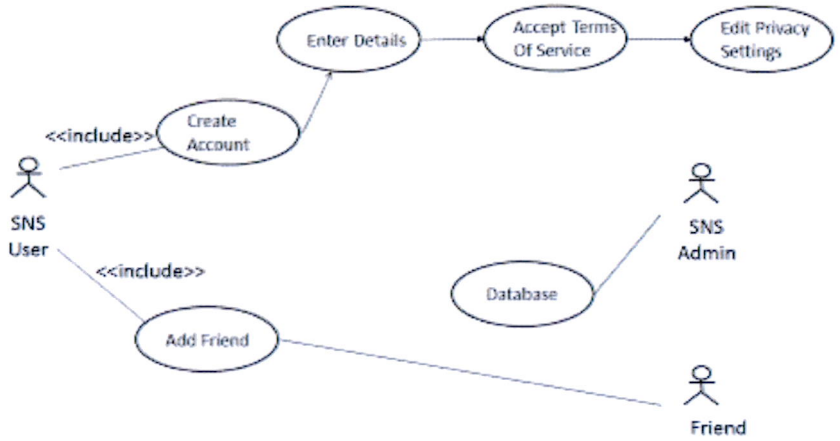
iii. The SNS shall allow each user to add a large number of friends

3.d Use cases are one way of capturing requirements. Give THREE advantages of use cases for capturing requirements and TWO disadvantages

[5 marks]

3.e The following is a use case diagram for the SNS. Copy the diagram into your answer book and clearly mark FOUR errors in the use case diagram. Explain the reason for each error.

[4 marks]



Question 3 continues on next page...

Question 3 continued...

3.f An automated car parking system controls traffic entering and leaving a car park. On entry, drivers take a ticket and an entry barrier is raised. Before returning to their car, drivers pre-pay for their parking time at a ticket machine. When they leave, drivers simply enter their (pre-paid) ticket and the exit barrier is raised. The system keeps track of the number of parked cars and does not allow new cars to enter when the car park is full. If there is a problem either at entry, exit or during payment, a car park attendant can be contacted for assistance.

Draw a use case diagram for the automated car parking system.

[8 marks]

Total 25 marks

Question 4

4.a A student information system is needed for students applying for, and studying on, a degree course. As part of the design, draw a statechart to model the changing status of a student as they progress from being an applicant to a graduate. Your statechart should include the following states and transitions between states:

- **states** = Applied, Accepted, Studying, Intercalating, Passed, Graduated, Failed, Withdrawn
- **events** = apply, accepted, register, pass, fail, resit, suspend, restart, graduate, withdraw

A failed student is able to resit to pass, but can only do so once.

A student can withdraw at any point after they have been accepted.

[12 marks]

4.b **Reuse** is a desirable quality of a software design. Describe what reuse means and why it is desirable.

[3 marks]

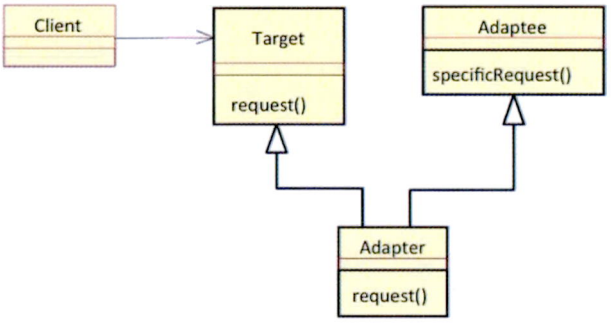
4.c What is a software **design pattern** and how may it contribute to the achievement of reuse in software?

[2 marks]

4.d Briefly explain the role of the **Adapter** design pattern.

[2 marks]

4.e Explain what is wrong with the UML class diagram below that attempts to represent the Adapter pattern using **Class Adaptation** and re-draw the diagram to correct the error.



[6 marks]

Total 25 marks

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